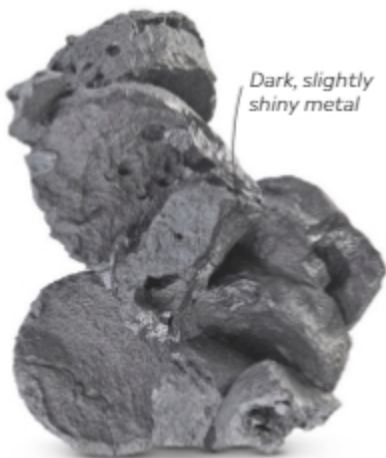


Boron

Dark, slightly shiny metal



Laboratory sample of pure boron

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B

In nature, boron is essential in small quantities for healthy plant growth. Some boron compounds are antiseptic and antifungal, used to treat minor wounds. This hard metalloid is used to make a type of heat-resistant glass called borosilicate. Boron is also added to fiberglass to strengthen it.

Ooey-gooey slime

Borax—a compound of boron, oxygen, and sodium—is a popular household cleaning chemical. It can also be used to make slime, or flubber—a rubbery play material that flows through fingers like a thick fluid but behaves like a solid when squeezed.

Bright color comes from food coloring additives.



Keeping food fresh

As a metal, aluminum is malleable, meaning that it can be fashioned into very thin foil-like sheets. This foil is used in the packaging of food and medications. With a thickness of just fractions of a millimeter, it provides a complete barrier to light, oxygen, moisture, and bacteria.

Heat-resistant foil



Recycling

Despite aluminum being the most widely available metal on Earth, extracting it is an expensive process. This is mainly due to the large amounts of electricity used to extract the metal from aluminum oxide, which has a high melting point. Recycling the metal—by making soft drink cans, for instance—is important for industry because reusing aluminum is 90 percent more energy-efficient than making new aluminum.



An average drink can contains about 70 percent recycled aluminum.

Aluminum fuselage is lightweight, so the plane uses less fuel to fly.



Transmitting electricity

Aluminum is an excellent conductor of electricity, and so it is an ideal metal for use in power cables. The light weight of the aluminum cables means that fewer pylons are required to support them, reducing the cost of construction and shortening the time it takes to build networks of cables.



Flying metal

Light, strong, and resistant to corrosion, aluminum is ideal for use in the manufacture of aircraft. A modern plane can contain up to 80 percent aluminum—used in its pure form as well as in many high-strength alloys.